

Project Title

Implement a system to compute the Longest Common Prefix (LCP) array from a suffix array. The application should determine the longest shared prefixes between adjacent suffixes and support substring-related queries. It should display both suffix and LCP arrays for visualization. The project should compare execution time with naïve approaches. This system is useful in compression and text analytics.

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

A.Y. 2026-2027

Submitted by

Parvathaneni Jaahnavi (2520030441)

Bhavani Tanya Sri (2520030112)

Noor UL Huda(2520030240)

Team Number: 12

Section: 5

Under the guidance of

D.Swathi

ASSISSTANT PROFESSOR, CSE



Hyderabad-500090, Telangana, India.

ABSTRACT

The proposed system implements an efficient method to compute the Longest Common Prefix (LCP) array from a given suffix array. The LCP array stores the length of the longest common prefix between every pair of adjacent suffixes arranged in lexicographical order. The system will generate the suffix array and calculate its corresponding LCP array, presenting both clearly for visualization and analysis. It will also support substring-related queries, helping users identify repeated patterns and common prefixes within a given text. To evaluate algorithmic efficiency, the system will compare the execution time of the optimized LCP computation method with a naïve approach that compares adjacent suffixes character by character. This comparison will demonstrate the advantages of efficient algorithms when processing larger strings. The proposed application can be useful in text processing, data compression, pattern matching, information retrieval, and text analytics, where fast identification of repeated or related substrings is important and useful.

Signature of the Guide